A partial derivative differentiates a function of multiple variables with respect to one variable, treating all other variables as constants.

From Foward Kinematics:

$$x = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2)$$

$$y = L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2)$$

differentiating x with respect to θ_1, treating θ_2 as constant:

The derivative of $\cos(\theta_1)$ with respect to θ_1 is $-\sin(\theta_1)$

The derivative of $\cos(\theta_1+\theta_2)$ with respect to θ_1 is $-\sin(\theta_1+\theta_2)$

$$\frac{\partial x}{\partial \theta_1} = -L_1 \sin(\theta_1) - L_2 \sin(\theta_1 + \theta_2)$$

Differentiating x with respect to θ_2, treating θ_1 as constant:

The derivative of $L_1 \cos(\theta_1)$ with respect to θ_2 is zero, because it does not contain θ_2

The derivative of $\cos(\theta_1+\theta_2)$ with respect to θ_2 is $-\sin(\theta_1+\theta_2)$

$$\frac{\partial x}{\partial \theta_2} = -L_2 \sin(\theta_1 + \theta_2)$$

Differentiating y with respect to θ_1, treating θ_2 as constant:

The derivative of $\sin(\theta_1)$ is $\cos(\theta_1)$

The derivative of $\sin(\theta_1+\theta_2)$ with respect to θ_1 is $\cos(\theta_1+\theta_2)$

$$\frac{\partial y}{\partial \theta_1} = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2)$$

Differentiating y with respect to θ_2, treating θ_1 as constant

The derivative of $L_1 \sin(\theta_1)$ with respect to θ_2 is zero, because it does not contain θ_2

The derivative of $\sin(\theta_1+\theta_2)$ with respect to θ_2 is $\cos(\theta_1+\theta_2)$

$$\frac{\partial y}{\partial \theta_2} = L_2 \cos(\theta_1 + \theta_2)$$

Assembling the Jacobian Matrix

The four partial derivatives are arranged in a 2x2 matrix, where each entry describes how one output (x or y) changes when one joint angle (θ_1 or θ_2) rotates:

$$J = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} \end{bmatrix} = \begin{bmatrix} -L_1 \sin\theta_1 - L_2 \sin(\theta_1+\theta_2) & -L_2 \sin(\theta_1+\theta_2) \\ L_1 \cos\theta_1 + L_2 \cos(\theta_1+\theta_2) & L_2 \cos(\theta_1+\theta_2) \end{bmatrix}$$

Since each column of J describes how the end effector move when one joint rotate alone, the total end-effector velocity ~~va~~ can be found by combining the two columns and weighting them by how fast each joint is ~~moving~~ rotating:

$$\begin{bmatrix} v_x \\ v_y \end{bmatrix} = J \begin{bmatrix} \dot\theta_1 \\ \dot\theta_2 \end{bmatrix}$$